

Work Package 3: Motion planning & robot control

The third work package consists of the automation tasks in which you will implement offline-computed minimum-jerk trajectories and an online motion planning approach to execute a pick-and-place workflow with the robot.

You are free to use your own gripper, but you may also use the standard gripper provided by the lecturers. As the gripper dimensions and principles vary between groups, you are free to place the cubes such that you may pick them with your gripper. Restrictions on cube placement are given below.

Before working on the actual hardware, validate your approaches in simulation.

Submission Guidelines:

- **Deliverables:**
 - o A ROS2 package containing the application
 - Package name: `rasc_l_wp3_ss26_group<Number>`
 - Node name: `wp3_tsk<1/2>`
 - Launch file name: `wp3_tsk<1/2>.launch...`
 - o Documentation on cube placement, choice of the radii – how did you determine the feasible region for your robot-gripper combination?
 - o The code used to produce the minimum-jerk trajectories. It must include the input and the output files.
 - o A video of the robot arm carrying out the two tasks based on your code. Be reminded that it must show the final stable stack in task 1 for at least 5 seconds. Only submit one video file that contains both tasks.
 - o Wrap everything in a zip folder and upload to ILIAS
- **Acceptance criteria for task 2:**
 - o You must fix a date with the tutors to take a validate video of task 2 during their tutor sessions. They are going to place the cubes at random positions and they are going to confirm the task 2 fulfillment to me.
 - o **Fix the date at least one week in advance directly with the respective tutor.**
 - o Task 2 must be included in the video
- Upload a new container containing everything we need to run your applications to your respective group folder in ILIAS

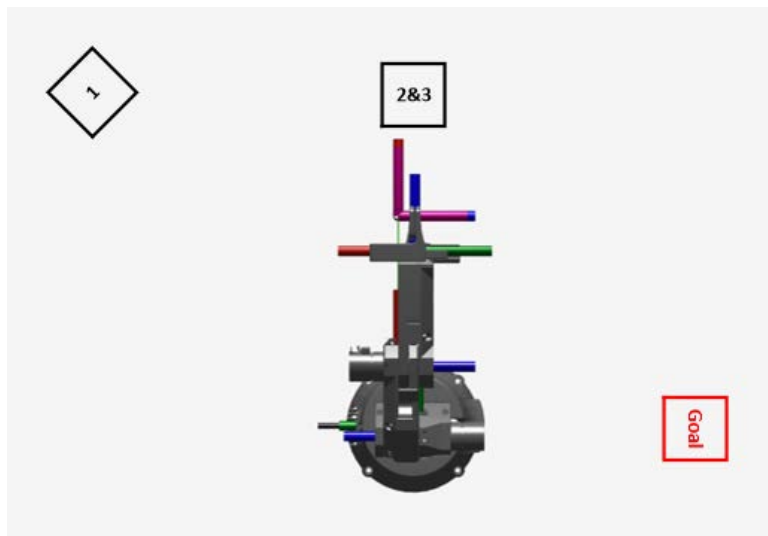
Submission deadline: Monday August 3, 23:59.

Task 1 – offline motion planning:

Pick three cubes and stack them one above the other. The task is deemed to be fulfilled when you successfully built a stable tower of three cubes. It must stand for at least 5 seconds.

Restrictions on cube placement:

- Two cubes must be stacked but they must be moved individually.
- Each cube must be moved to a new place.
- Every stack must be placed on a different radius:
 - Cube 1 with the largest, the goal position with the smallest radius. Cubes 2&3 in between.
- Cube placement must be placed as shown in the picture below:
 - Cube 1 must be placed at left of the robot's reference switch.
 - The cubes 2 and 3 are near the reference switch. Cubes 2 is at the bottom and cube 3 is on top.
 - The goal position is on the right side.
 - All three cube positions must differ significantly regarding the radius.



- The cubes must be stacked in order 1-2-3 with cube 1 at the bottom, cube 2 in the middle, and cube 3 on top.

You are required to solve the task in the following way:

- Pre-compute minimum-jerk trajectories to move the cubes. Every major robot movement has to be based on such a trajectory
- Your node shall accept these offline-trajectories and execute the movement
- You are required to use the motion controllers CSP mode
- There are no further restrictions on how you implement your solutions.

Task 2 – Online motion planning:

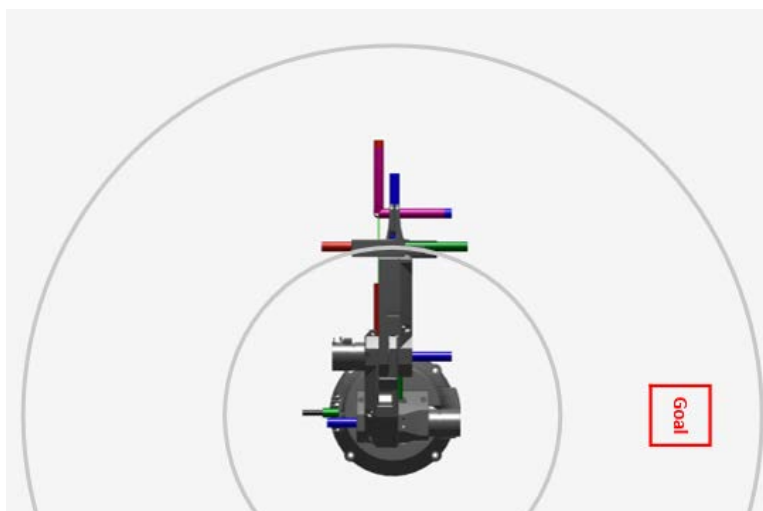
Pick an arbitrary number of cubes and place them in the same goal each time.

The position of the cubes is not known in advance, the goal position is known. The cubes center position is published during runtime by the user to the topic `/goal_poses` via the message type `geometry_msgs/msg/Point`. In this task, only one cube is present in the system at a time, the goal position is always empty.

Constraints regarding cube placement:

- Cubes will be placed during the runtime of your node.
- The cubes will always be placed such that they are oriented with their faces in parallel to the grippers' contact faces, according to the box plate.
- You are required to provide the feasible region of your gripper defined by the minimum and the maximum feasible radius from the robot base. Feasible means that no self- or world collisions occur and that the cube is still within reach of the gripper.
 - You are not allowed to restrict your robot systems' feasible region for any other reason than the above-mentioned kinematic considerations. Doing so results in failure of the task.
- The radius of the target must be your robot's maximum feasible radius.
- The target is placed on the right side of the robot. Cubes may be placed anywhere between the feasible radii and a shoulder joint angle between $+0.5\pi$ to -0.5π . Pick and place both happens on the box plate.

No further constraints to your solution apply.



Hints and Information

Literature:

- Lynch and Park: Modern Robotics, 2017. Free pre-print version online
- <https://manipulation.mit.edu/>
- Springer Handbook on Robotics. Available via the KIT-library.
- Minimum-jerk-trajectory generation:
 - o Scientific literature mentioned in the intro slides
 - o Mathworks online
 - o Do your own research as this is a central part of this work package

Videos of applications for the interested

- Online motion planning in pick-and-place in parcel sorting:
<https://youtu.be/ovcPNAwiFIM?si=e6qOyuBhGcEUQ5ea&t=44>
<https://youtu.be/Fs6XvQqDBQg?si=hG0BREJlwWfTIOR7>
The second automation task is inspired by this application. Picking from various, previously unknown positions and, at least, as shown in the second video, placing at a previously known target. However, as shown in the first video, there may be additional constraints such as orientation constraints and/or moving targets.
- Video summary of a paper on the benefits of jerk-reduction:
<https://www.youtube.com/watch?v=9p0bmgBnL2Q>